

# Analyzing the Impact of COVID-19 on U.S. Employment Through Data Visualization

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**Abstract**—Millions of people have been impacted by the COVID-19 pandemic drastically. This pandemic has challenged different areas of people's lives, such as employment, health, the economy, education, etc. Employment rate is a significant factor in economic growth, poverty reduction, and social stability. Given the significant impact of the pandemic since early 2020, it is crucial to examine how it has affected employment rates. In this paper, we utilize data visualization to present a comprehensive analysis that can aid government efforts in planning and managing employment during pandemics such as COVID-19. In this work, we use the collected data from the New York Times COVID-19 dataset and data from the U.S. Bureau of Labor Statistics.

**Index Terms**—R, Data Visualization, COVID-19, Employment Rate

## I. INTRODUCTION

At the start of 1918, when the Spanish flu pandemic began, World War I had just ended, with an estimated 40 million military and civilian casualties. Near the end of World War I, reports of the pandemic were kept quiet by the censors of the warring nations in order to maintain morale. Approximately 500 million people, or one third of the world's population, were infected in four waves over a period of two years. The number of fatalities is estimated to be between 17 million and 50 million.

At the beginning of 2020, the world was struck by another pandemic, COVID-19. The main distinction between the 1918 and 2020 pandemics is that there was no global conflict occurring at the time of the latter. Also since the beginning of the last century, there has been a tremendous amount of progress in medical and virology sciences, healthcare, and other fields. Considering the advancements that have been made, it was unexpected that the world was not prepared for another pandemic like the Spanish flu but to everyone's surprise, we were not. Corona viruses, which are enveloped, single-stranded positive-sense RNA viruses, possess some of the largest viral genomes, approximately 32 kilobases in length. These viruses can infect both humans and a broad spectrum of animals [1]. The COVID-19 pandemic has had a drastic effect on millions of people including employment. This study employs data visualization to evaluate the effect of COVID-19 on the unemployment rate in the United States.

Numerous researchers have examined the impact of COVID-19 on public transit demand and unemployment insurance [2], [3]. However, a thorough analysis of the relationship between COVID-19 and employment remains lacking.

This paper analyzes COVID-19 based on the available data from New York Times, U.S Bureau of Labor Statistics [4], [5]. The analysis covers various aspects of COVID-19, including case numbers, death rates, and employment impacts. Additionally, we present data visualizations comparing employment across different U.S. states in relation to their COVID-19 case numbers. In this work we used R programming language citerlanguage. R is a tool for performing statistical operations and creating visualizations. It offers a variety of features and methods to create graphical representations of data.

The remainder of the paper is organized as follows: Section II-A reviews the research conducted in the field of data visualization and its applications in addressing pandemic-related issues. Section II-B details the tools used for data visualization, with a focus on the R programming language. Section III showcases the visualizations depicting the number of cases and deaths since the onset of the pandemic in 2020 and Section IV provides concluding remarks.

## II. RELATED WORK

### A. Data Visualization

The history of data visualization technology can be traced to 1950s computer graphics, where computers were first used to create charts. Data visualization technology has advanced steadily, leading to its widespread use across a variety of industries, including e-commerce, meteorology, education and others. Researchers have extensively studied this technology and its applications. Xiangdong *et al.* utilized thermal maps through data visualization technology to visualize the operation data related to tourism e-commerce, and based on the results, they proposed targeted suggestions for e-commerce product promotion strategies [6]. Visualization techniques are also utilized in [7], leveraging individual customer load profiles alongside clustering classification as a methodology.

In other work, Afzal *et al.* [8] developed visual analytics environment that aids in modeling, simulating, and exploring COVID-19 spread using county-level data such as demographics and hospital capacity. Several studies and works focused

on visual analytics and visualization have been presented in recent years for analyzing epidemic or pandemic data [8]–[10].

### B. Tools and Datasets for Visualization

There are numerous data visualization tools due to the growth and utilization of data visualization technology. Fundamentally, data visualization technology transforms abstract information conveyed through numbers, texts, and other data into visual representations. Indeed, data visualization analysis encompasses a comprehensive series of processes, starting from data acquisition and preprocessing to visual mapping and human-computer interaction. Given its critical role in epidemic prevention and control, the data visualization analysis process must be conducted with greater rigor [11].

Today's data visualization tools span a broad spectrum, catering to enterprises, developers, and general users alike. Researchers are increasingly adopting the programming language R [12] for data processing and statistical analysis, as evidenced by an average annual growth rate of 87% [13]. Besides enhancing reproducibility and transparency, one of the advantages of using R is that researchers can access a much broader array of fully customizable data visualization options compared to what is typically available in point-and-click software, thanks to R's open-source nature. These visualization choices not only enhance visual appeal but also promote transparency by revealing the distribution of underlying data, moving beyond commonly used aggregated visualizations. In addition to enhancing reproducibility, utilizing R for data visualization provides researchers with nearly complete control over each element of the plot [14].

## III. VISUALIZATION ANALYSIS OF EPIDEMIC DATA

### A. U.S COVID-19 Cases and Death Tolls

At the start of COVID-19 pandemic, scientists determined that the best approach to stop the spread of the Corona Virus was to implement a public lockdown and put an immediate stop to day-to-day activities for everyone except essential workers. After the lock down, many people were able to transition from office work to remote work and use tools such as Zoom to remain employed. Despite this, many employees from other sectors, such as manual laborers, were unable to continue work. At the beginning of the pandemic, when lockdowns were put in place and the fear of the virus was high, those who had lost their jobs faced the toughest times.

As the pandemic continued, employers looked for ways to adjust their work environments for staff and observe social distancing, including the use of face masks and other strategies to reduce the transmission of the virus. This was the pivotal moment and we noticed that unemployment rate decreased compared to start of 2020. The unemployment rate was still higher than 2019. Examining the data further in Figure 3, it is evident that 2021 was the most difficult year for employees and the job market in the United States. This could be due to the workers' fear and anxiety of becoming ill in workplace and potentially losing their lives. After the U.S. developed an

mRNA vaccine in 2022 and began mass immunization, the workforce regained hope of regaining lost jobs and the unemployment rate decreased. In fact the lowest Unemployment rate among the pandemic years is in 2022.

This section presents the visualization of number of cases and number of deaths since the start of pandemic in 2020. Figure 1 shows the number of new cases and deaths per month. Figure 2 shows the cumulative cases and death percentage.

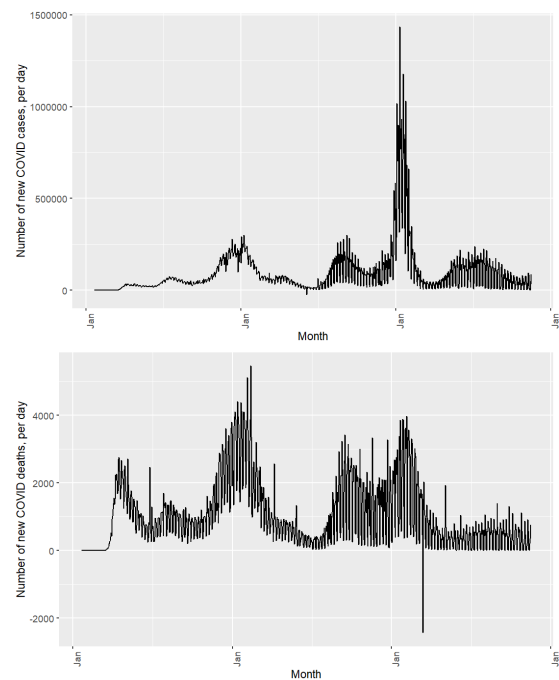


Fig. 1: Illustrating the COVID-19 new cases and death per month.

Figure 1 shows the number of cases and fatalities began to increase in April. As a result, the public shutdown started during this period. As comprehensive data shown in Figure 1, the death toll were started to rise in April 2020 to almost 3000 death per day and peaked in February 2021. Based on Figure 2 number of new cases was less than 250,000 in year 2020 and 2021. This death toll started to decrease until Fall 2021. At the end of 2021 and beginning of 2022 the death toll quickly peaked again and started to slow down after February 2022 to nearly 1,000 death per day. In April 19, 2021, vaccine roll out began and all U.S. states have made it possible for people aged 16 and above to receive the vaccine. This suggests that the initial drop in fatalities could be linked to getting vaccinated.

After the second dose of the vaccine was made available to the public after a 6-month period, it was expected that the death toll would decrease; however, the data indicates otherwise. This could be due to the emergence of new strains, the easing of social distancing measures, or people's reluctance to get vaccinated. However further investigation and analysis is required. Analyzing the data shows the minimum number of cases was in 2020-01-21 with one case and maximum in 2022-11-15 with 97842103 cases and mean in 2021-03-02 with 40931899 cases. The statistics of the death tolls shows

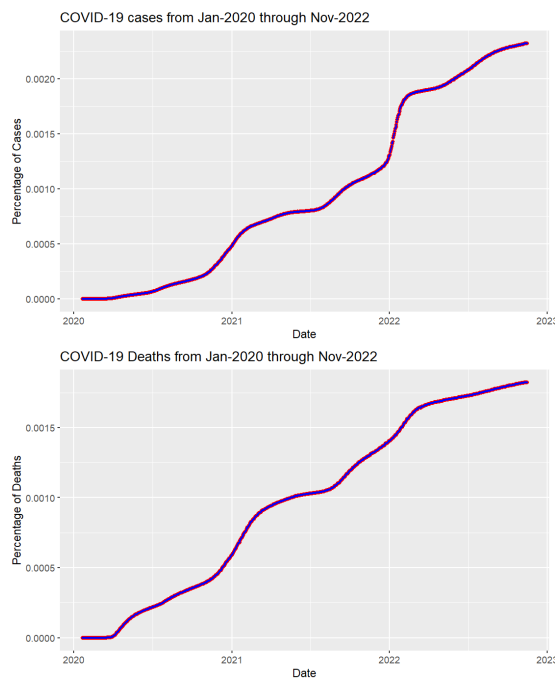


Fig. 2: Illustrating the COVID-19 cumulative cases and death percentage.

minimum number of deaths was in 2020-01-21 with no case and maximum in 2022-11-15 with 1070970 deaths and mean in 2021-06-18 with 570441 deaths.

### B. Unemployment and Other Various Rates in U.S

In this section we present the visualization results for unemployment data from U.S Bureau of Labor Statistics from January 2020 to October 2022. Figure 3 is a comprehensive visualization results that shows different measures of unemployment from January 2020 to October 2022. Figure 4

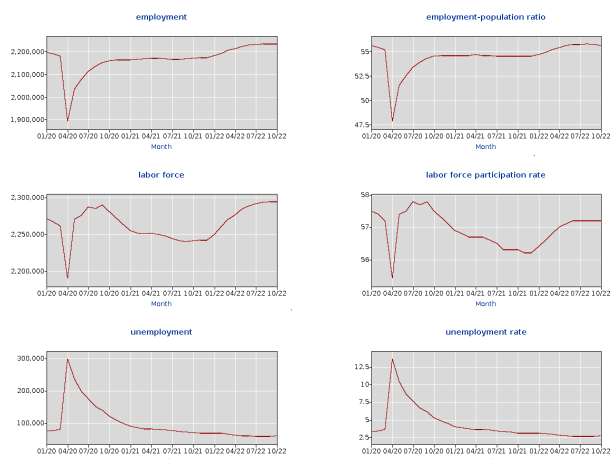


Fig. 3: Illustrating various unemployment measures from January 2020 to October 2022.

includes maps that show the changes in unemployment rate for U.S states from 2020 to 2022.

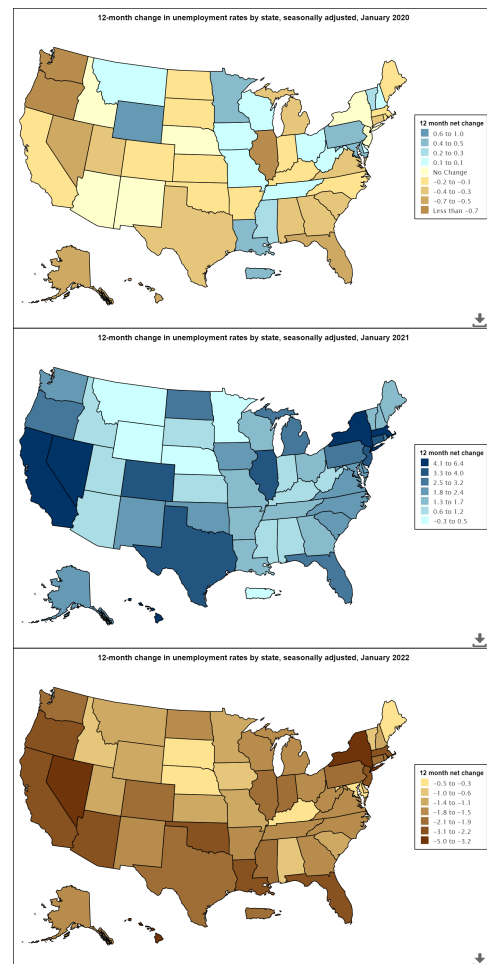


Fig. 4: Illustrating the U.S unemployment maps from 2020 to 2022.

As mentioned in III-A, Figure 1 shows new COVID-19 cases were less than 250,000 in year 2020 and 2021. Towards the end of 2021, this number began to increase. The highest amount of cases was recorded in the winter of 2022, reaching almost 1,500,000 per day. Figure 4 visualizes the unemployment rate percent for each state during the pandemic. The Figure shows in the year 2020 less than half of states had noticeable changes in employment rate. In 2021, many states have been impacted by an increase in unemployment rates compared to the previous year, ranging from 0.6 to 6.4. In 2022, the majority of US states have experienced change in their unemployment rate. This indicates by initiating the vaccination process and easing pandemic regulations, people began to return to their jobs. As the vaccine rollout and social restrictions were eased, unemployment rates in most states dropped from -0.5 to -3.2, showing that the workforce was beginning to find employment. The results show the job market experienced almost the same unemployment rate compared to the year 2020 before the lockdown.

Based on news sources and economic data from the World Bank, it is evident that the US unemployment rate is lower than other nations, even in the face of the COVID-19 pan-

demic. But unemployment rate was adversely affected during pandemic due to a variety of reasons such as reduced shopping in stores, lack of access to social events like concerts and exhibitions, etc. Figure 5 presents visualization results for unemployment rate per states to measure the COVID-19 impact on each state. It shows unemployment rate, new COVID-19 cases and number of deaths per state from 2020 to 2022.

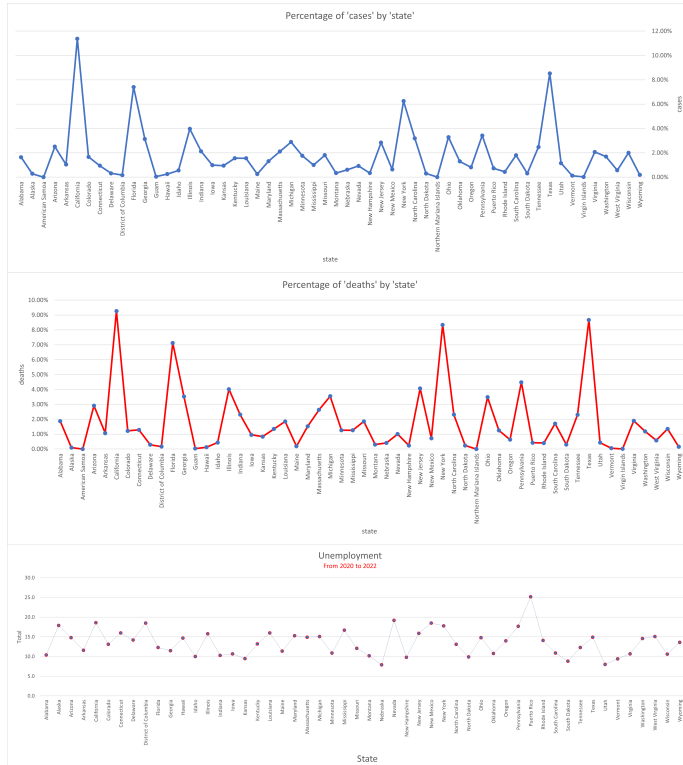


Fig. 5: Illustrating the various visualization results.

### C. Comparative Analysis

In this section we present comparative visualization result in order to provide a comparative analysis. Figure 6 shows unemployment rate, new COVID-19 cases and number of deaths per state in one graph. Figure 6 shows that case

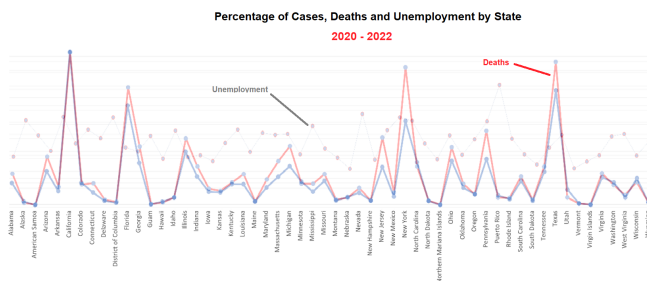


Fig. 6: Illustrating comparative visualization results.

percentage are higher in states such as California, Texas, Florida and New York. Death tolls are also higher in states such California, New York, Texas and Florida. Results also

indicate unemployment rate is higher for states such as Puerto Rico, District of Colombia and Nevada. Results show that although there is a correlation between the number of new cases and deaths, it is not linked to the rate of unemployment. The higher population in California, New York, Texas and Florida is leading to a higher rate of infection and death due to the concentration of people in these areas. The unemployment rate was greater in areas with smaller populations such as Puerto Rico, Nevada, and the District of Columbia, which could be attributed to the variety of jobs available in these different states. In April, after governments shut down the economy, the unemployment rate reached 14.7%, the highest since the great depression. Ultimately, due to the measures implemented by the government [15], [16], unemployment rates improved. The U.S. unemployment rate was 3.7% for November 2022 which is much less than April 2022.

### IV. CONCLUSION

This study offers data analysis and visualization to illustrate the varying impacts of COVID-19 on employment across all U.S. states. The COVID-19 pandemic has had a tremendous effect on millions of people, altering their lives in many ways, including employment, health, economy, and education. Despite other factors that can lead to joblessness, the COVID pandemic has drastically increased the rate of unemployment. Using data visualization techniques we assessed the impact and connection between the number of new cases, fatalities and unemployment rate.

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